

Unit 1.8 Inverse functions

Note Title

2012/02/22

Stewart, Section 1.6 p 58-62

- Not all functions possess inverses
- If f never takes on the same value twice we say it's one-to-one.

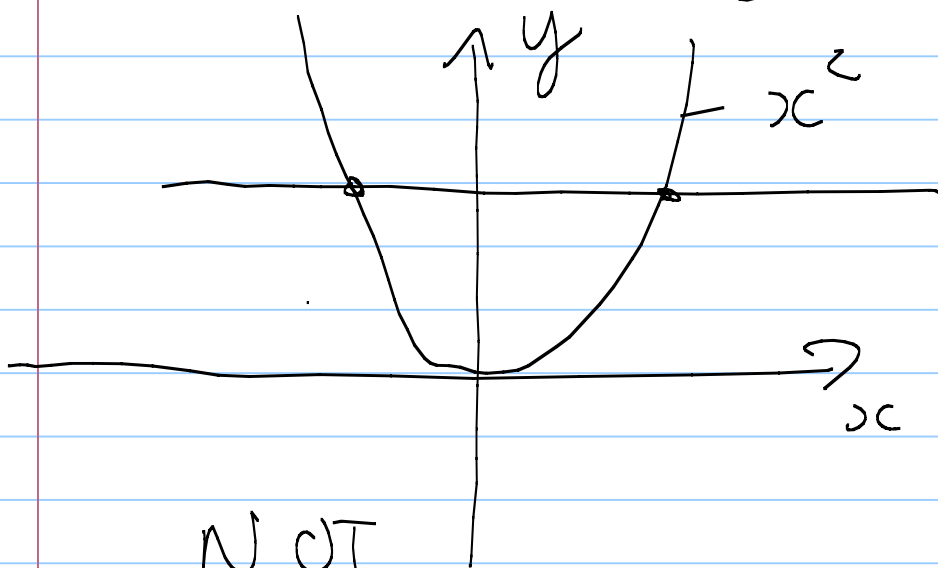
Definition

- f is a one-to-one function if $x_1 \neq x_2 \Rightarrow f(x_1) \neq f(x_2)$
- ^{or} f is a one-to-one function if $f(x_1) = f(x_2) \Rightarrow x_1 = x_2$

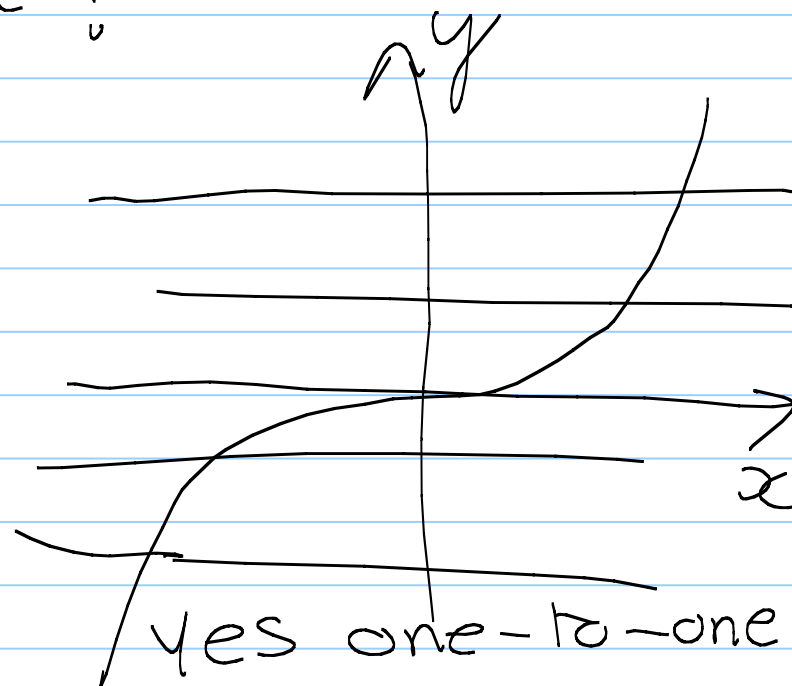
Horizontal line test

A function is one-to-one if no horizontal line intersects its graph more than once.

Example. Is $f(x) = x^2$ and $g(x) = x^3$
one-to-one?



NOT
one-to-one



yes one-to-one

Definition

Let f be a one-to-one function with domain A & range B .

Then its inverse function f^{-1} has a domain B and Range A and is defined by

$$f^{-1}(y) = x \iff f(x) = y$$

for any y in B .

domain of f^{-1} = range of f .
range of f^{-1} = domain of f .

Example If $f(1) = 5$, then
 $f^{-1}(5) = 1$.

- The letter x is traditionally used as the independent variable. So we write:

$$f^{-1}(x) = y \iff f(y) = x$$

The cancellation equations

Let $f: A \rightarrow B$ with B the range of f and f is one-to-one.

Then $f^{-1}: B \rightarrow A$ and

$$(f^{-1} \circ f)(x) = f^{-1}(f(x)) = x \text{ for all } x \in A$$

and

$$(f \circ f^{-1})(x) = f(f^{-1}(x)) = x \text{ for all } x \in B.$$

How to find The inverse of a One-to-One function f .

① $y = f(x)$

② Solve for x in terms of y

③ interchange x & y .

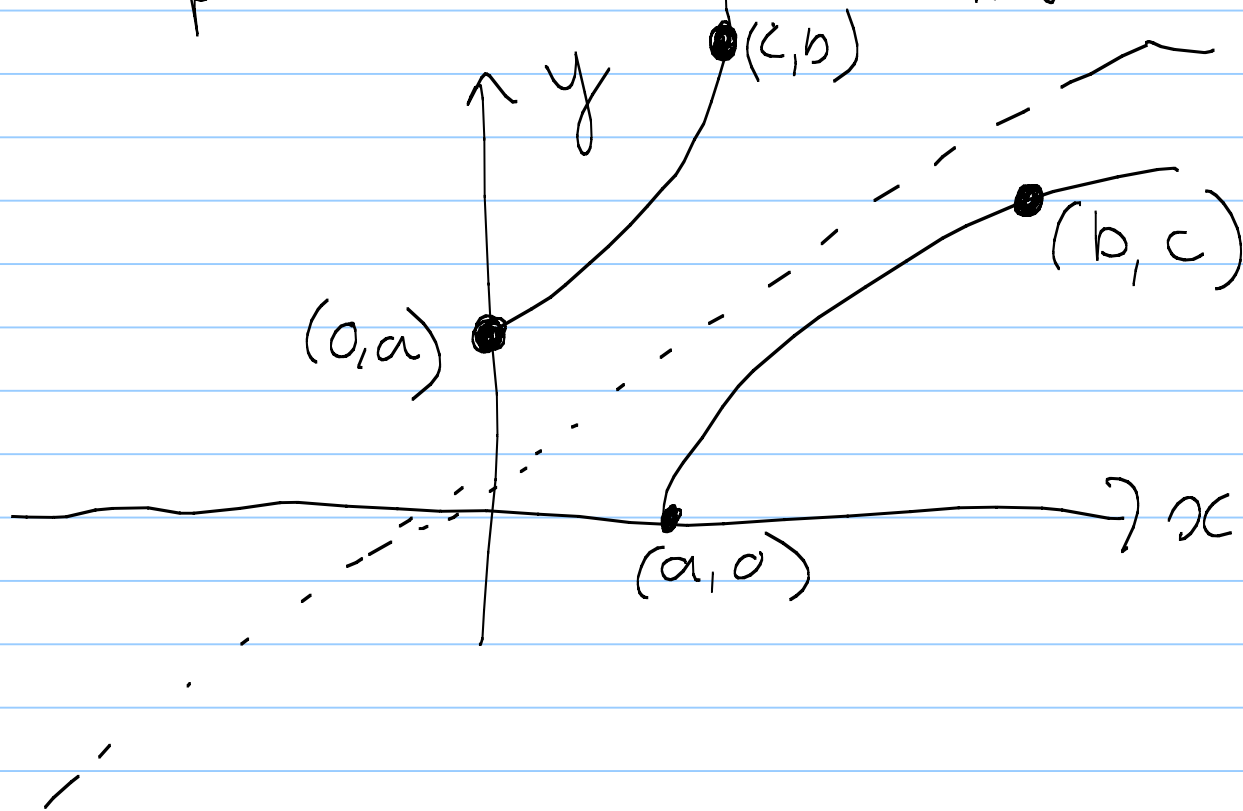
resulting equation is $y = f^{-1}(x)$.

The graph of f^{-1} is obtained by reflecting f about the line $y = x$.

Example

Sketch

inverse $y=x$.



Example Find The inverse

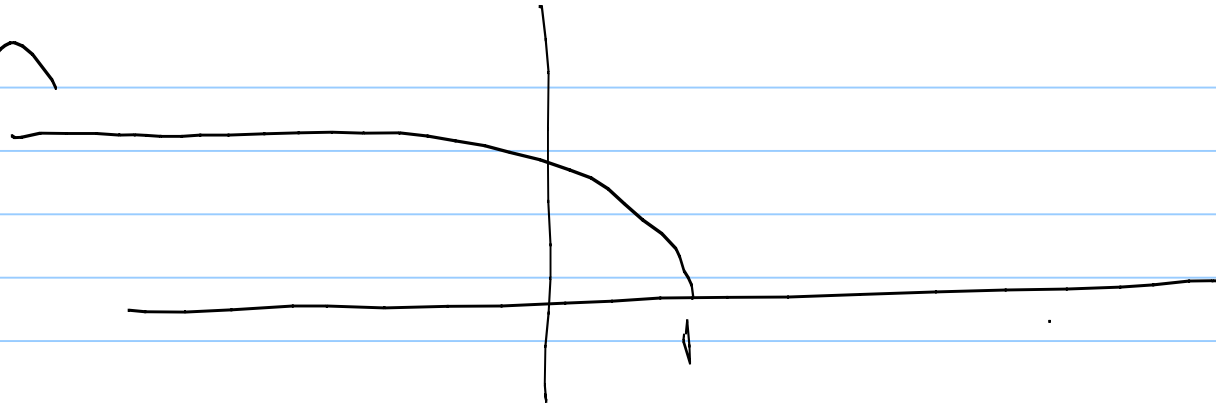
1. $f(x) = \sqrt{1-x}$

The Domain $1-x \geq 0$

$$\Rightarrow 1 \geq x$$

$$\Rightarrow x \in (-\infty, 1]$$

Sketch



The function is one-to-one.
and

$$f: (-\infty, 1] \rightarrow [0, \infty)$$

Finding the inverse:

$$y = \sqrt{1-x}$$

$$\Rightarrow y^2 = 1-x$$

$$\Rightarrow x = 1 - y^2$$

$$\Rightarrow y = \sqrt{1-x}$$

$$f^{-1}(x) = \sqrt{1-x}$$

and $x \in [0, 1]$ and $y \in (-\infty, 1]$

Example Find inverse

$$f(x) = \frac{x+4}{2x-5}$$

$$y = \frac{x+4}{2x-5}$$

$$\Rightarrow y(2x-5) = x+4$$

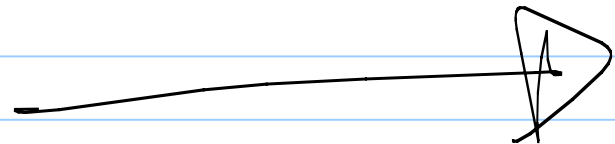
$$\Rightarrow 2xy - 5y = x+4$$

$$\Rightarrow 2xy - x = 4 + 5y$$

$$\Rightarrow x(2y-1) = 4+5y$$

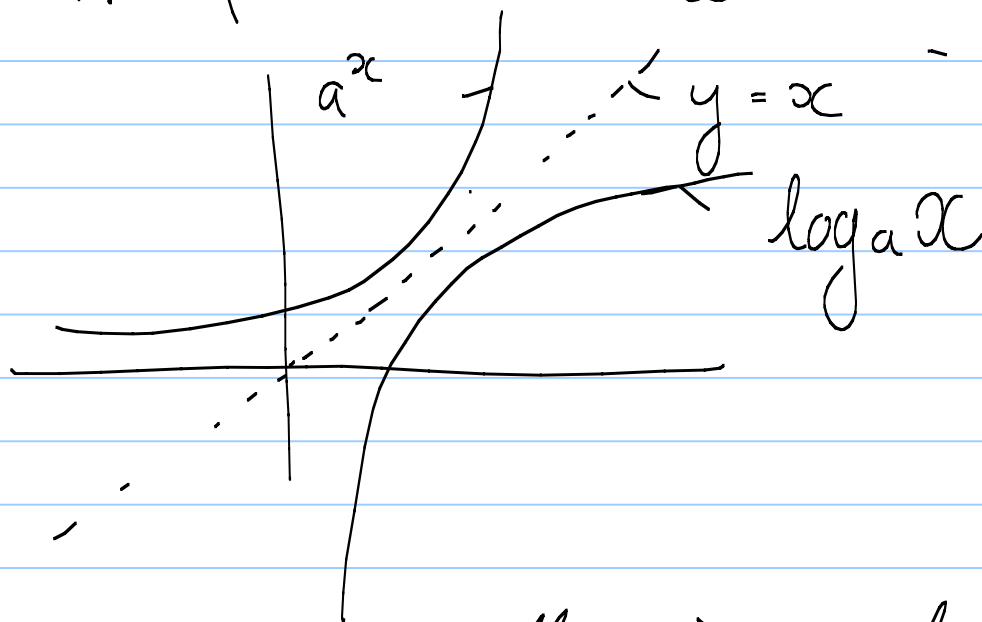
$$\Rightarrow x = \frac{4+5y}{2y-1}$$

$$\Rightarrow f^{-1}(x) = \frac{4+5x}{2x-1}$$



Logarithmic functions

- The function $f(x) = a^x$ has an inverse function



$f^{-1}(x) = \log_a x$ is called the logarithmic function with base a .

Natural Logarithms. . pg 64

- $f(x) = e^x$ is one-to-one and has an inverse

$$f^{-1}(x) = \log_e x = \ln x$$

Must use notation $\ln$ NOT $\log_e$

Definition

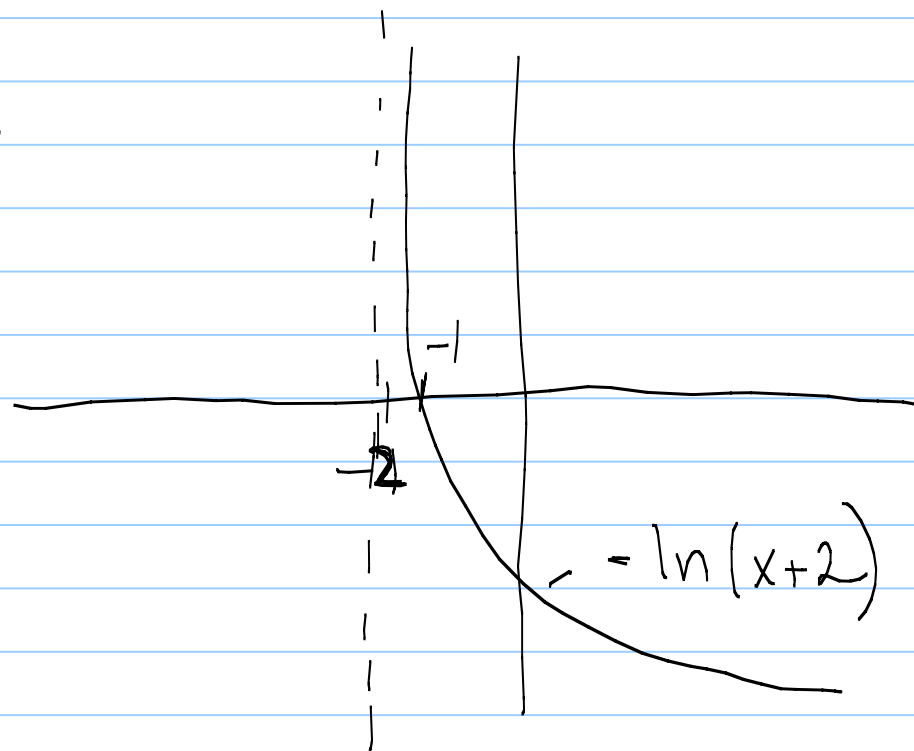
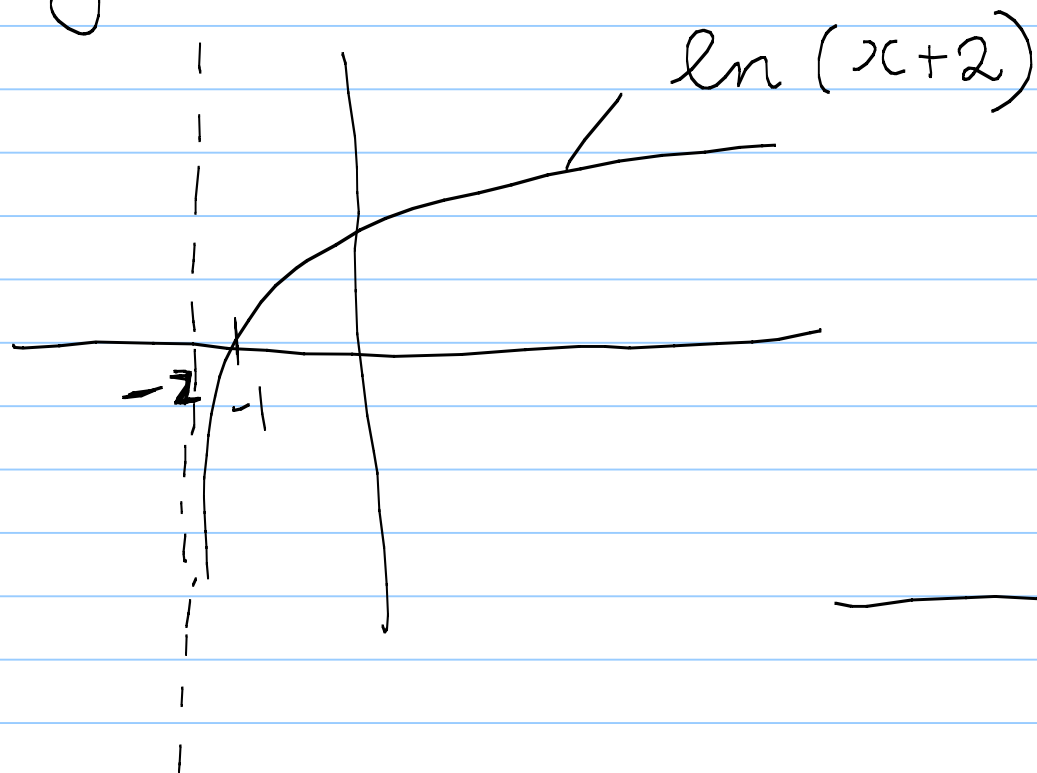
$$y = \ln x \iff e^y = x$$

Example: Sketch

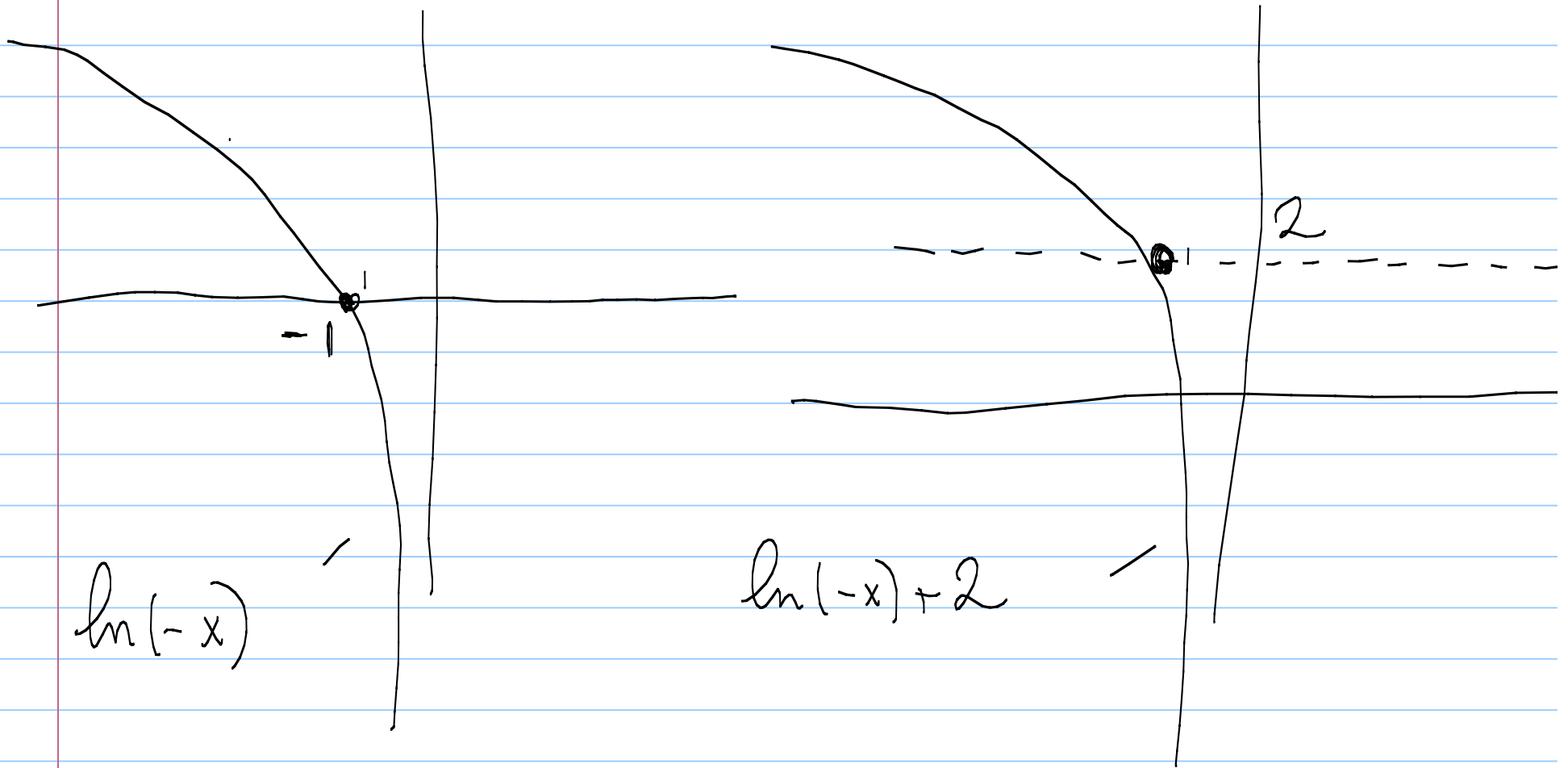
1) $y = -\ln(x+2)$

2) $f(x) = \ln(-x) + 2$

$$y = -\ln(x+2)$$



$$y = \ln(-x) + 2$$



(Memorize it)

Properties The $\ln$ -function

$$1] \ln(ab) = \ln a + \ln b, \quad a > 0, b > 0$$

$$2] \ln\left(\frac{a}{b}\right) = \ln a - \ln b, \quad a > 0, b > 0$$

$$3] \ln(a^b) = b \ln a, \quad a > 0$$

Cancellation Properties NB

$$\ln(e^x) = x \quad \text{for } x \in \mathbb{R}$$

$$e^{\ln x} = x \quad \text{for } x > 0$$

$$\ln e = 1$$

Example Solve equalities

$$1] e^{5x} = 4$$

$$2] \ln 2x = 8$$

$$3] \ln 20x = 4 + 3 \ln x$$

$$4] e^x = 9 e^{3x}$$

$$5) \quad 2 \ln x + \ln 4 = 3$$

$$6) \quad \frac{e^{-x} + 1}{e^{-x} - 1} = 2$$

$$7) \quad \frac{3 \ln x}{1 + \ln \sqrt{x}} = 4$$

$$8) \quad 7a^x = b^x$$

$$9) \quad e^{5x} = 4e^{20c}$$

Solutions

$$1) e^{5x} = 4$$

$$\Rightarrow \ln(e^{5x}) = \ln 4$$

$$\Rightarrow 5x = \ln 4$$

$$\Rightarrow x = \ln 4 / 5$$

$$2) \ln 20x = 8$$

$$\Rightarrow e^{\ln 20x} = e^8$$

$$\Rightarrow 20x = e^8 \Rightarrow x = \frac{e^8}{20}$$

$$3) \ln 20x = 4 + 3 \ln x$$

$$\Rightarrow \ln 20x = 4 + \ln x^3$$

$$\Rightarrow \ln x^3 - \ln 20x = -4$$

$$\Rightarrow \ln \left(\frac{x^3}{20x} \right) = -4$$

$$\Rightarrow \ln \left(\frac{x^2}{2} \right) = -4$$

$$\Rightarrow e^{\ln(x^2/2)} = e^{-4}$$

$$\Rightarrow \frac{x^2}{2} = e^{-4}$$

$$\Rightarrow x^2 = 2e^{-4}$$

$$\Rightarrow x = \sqrt{2e^{-4}}$$

Since $x > 0$

$$4) e^x = 9e^{3x}$$

Method 1

$$e^x = 9e^{3x}$$

$$\Rightarrow \frac{e^x}{e^{3x}} = 9$$

$$\Rightarrow e^{-2x} = 9$$

$$\Rightarrow \ln(e^{-2x}) = \ln 9$$

$$\Rightarrow -2x = \ln 9$$

$$\Rightarrow x = -\ln 9 / 2$$

OK method 2

$$e^x = 9e^{3x}$$

$$\Rightarrow \ln e^x = \ln (9e^{3x})$$

$$\Rightarrow \ln e^x = \ln 9 + \ln e^{3x}$$

$$\Rightarrow x = \ln 9 + 3x$$

$$\Rightarrow -2x = \ln 9$$

$$\Rightarrow x = \ln 9 / -2$$

$$5) \quad 2 \ln x + \ln 4 = 3$$

$$\Rightarrow \ln x^2 = 3 - \ln 4$$

$$\Rightarrow e^{\ln x^2} = e^{3 - \ln 4}$$

$$\Rightarrow x^2 = e^3 e^{\ln 4^{-1}}$$

$$= e^3 / 4$$

$$\Rightarrow x = \sqrt{e^3 / 4}$$

$$6) \frac{e^{-x} + 1}{e^{-x} - 1} = 2$$

$$\Rightarrow e^{-x} + 1 = 2(e^{-x} - 1)$$

$$\Rightarrow e^{-x} + 1 = 2e^{-x} - 2$$

$$\Rightarrow e^{-x} - 2e^{-x} = -3$$

$$\Rightarrow -e^{-x} = -3$$

$$\Rightarrow e^{-x} = 3$$

$$\Rightarrow -x = \ln 3$$

$$\Rightarrow x = -\ln 3$$

$$7) \frac{3 \ln x}{1 + \ln \sqrt{x}} = 4$$

$$\Rightarrow 3 \ln x = 4 + 4 \ln \sqrt{x}$$

$$\Rightarrow \ln x^3 = 4 + \ln x^{4/2}$$

$$\Rightarrow \ln x^3 - \ln x^2 = 4$$

$$\Rightarrow \ln \left(\frac{x^3}{x^2} \right) = 4$$

$$x = e^4$$

$$8) \quad 7a^x = b^x$$

$$\Rightarrow \ln(7a^x) = \ln(b^x)$$

$$\Rightarrow \ln 7 + \ln a^x = \ln b^x$$

$$\Rightarrow \ln 7 + x \ln a = x \ln b$$

$$\Rightarrow x [\ln b - \ln a] = \ln 7$$

$$\Rightarrow x = \frac{\ln 7}{\ln b - \ln a}$$

$$9) e^{5x} = 4 e^{2x}$$

$$\Rightarrow \ln e^{5x} = \ln (4 e^{2x})$$

$$\Rightarrow 5x = \ln 4 + \ln e^{2x}$$

$$\Rightarrow 5x - 2x = \ln 4$$

$$\Rightarrow 3x = \ln 4$$

$$\Rightarrow x = \frac{\ln 4}{3}$$

Example Solve inequalities

$$\boxed{1} \quad \ln x < 3$$

$$\Rightarrow e^{\ln x} < e^3$$

$$\Rightarrow x < e^3 \quad \text{AND} \quad x > 0$$

Since e is an increasing function

$$\Rightarrow x \in (0, e^3)$$

Homework solve for x

$$* \frac{\ln(5-2x)}{\ln x} = 1$$

$$* 2 \ln x = \ln(x^3) + 2$$

Example Find the inverse of

1] $f(x) = e^{-2x}$

2] $f(x) = \ln(2 + \ln x)$

3] $f(x) = e^{\frac{20x-3}{-7}}$

4] $f(x) = \frac{1 + e^x}{1 - e^x}$

$$1) y = e^{-2x}$$

$$\Rightarrow \ln y = \ln e^{-2x}$$

$$\Rightarrow \ln y = -2x$$

$$\Rightarrow x = \ln y / -2$$

$$\Rightarrow f^{-1}(x) = \ln x / -2$$

$$2) f(x) = \ln(2 + \ln x)$$

$$\Rightarrow y = \ln(2 + \ln x)$$

$$\Rightarrow e^y = e^{\ln(2 + \ln x)}$$

$$\Rightarrow e^y = 2 + \ln x$$

$$\Rightarrow e^y - 2 = \ln x$$

$$\Rightarrow e^{e^y - 2} = e^{\ln x}$$

$$\Rightarrow x = e^{e^y - 2}$$

$$\Rightarrow f^{-1}(x) = e^{e^x} e^{-2}$$

$$3) f(x) = e^{2x-3} - 7$$

$$\Rightarrow y = e^{2x-3} - 7$$

$$\Rightarrow y + 7 = e^{2x-3}$$

$$\Rightarrow \ln(y+7) = \ln e^{2x-3}$$

$$\Rightarrow \ln(y+7) = 2x-3$$

$$\Rightarrow \ln(y+7) + 3 = 2x$$

$$\Rightarrow f^{-1}(x) = \frac{1}{2} (\ln(x+7) + 3)$$

$$4) \quad y = \frac{1 + e^x}{1 - e^x}$$

$$\Rightarrow y(1 - e^x) = 1 + e^x$$

$$\Rightarrow y - ye^x = 1 + e^x$$

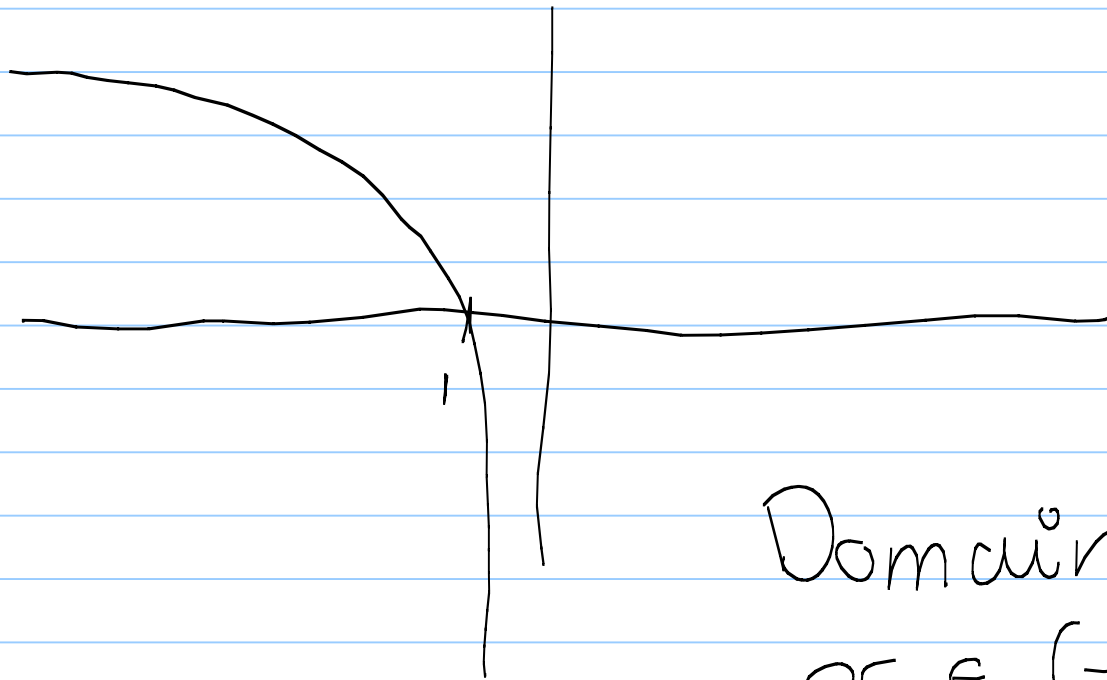
$$\Rightarrow y - 1 = e^x [1 + y]$$

$$\Rightarrow e^x = \frac{y - 1}{1 + y}$$

$$\Rightarrow f^{-1}(x) = \ln \left(\frac{x - 1}{1 + x} \right)$$

Example What is the Domain of

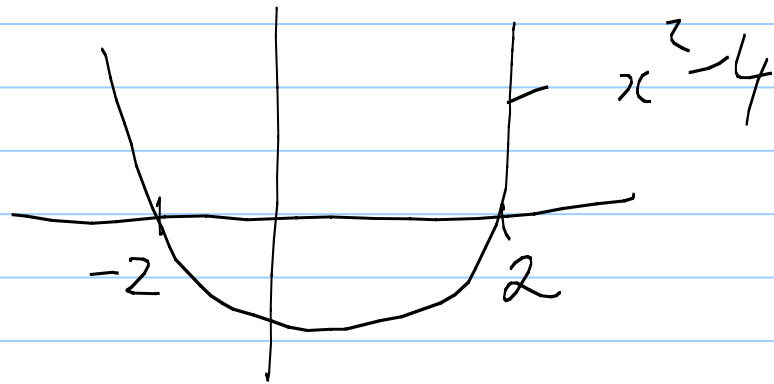
$$\square f(x) = \ln(-x)$$



Domain
 $x \in (-\infty, 0)$

$$2] f(x) = \ln(x^2 - 4)$$

$$\Rightarrow x^2 - 4 > 0$$



$$x \in (-\infty, -2) \cup (2, \infty)$$

